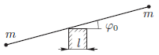
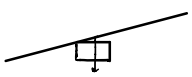
1. (40分) 37室

(改编自俄罗斯物理竞赛)如图有一根长度为L质量为m的均匀质量杆.杆中点放置在长度为$l \ll L$的固定底座中央。可以认为底座只有左右两个端点和杆能接触，杆在底座上即不滑动也不掉下来.初态将杆绕左端点转一个小角度φ_0，然后静止释放，此后杆在图中平面内运动.

(1) 假设碰撞为完全弹性碰撞，求来回运动的周期.

(2) 如果碰撞的恢复系数为$e = 1 - \delta$，其中$\delta \ll 1$，恢复系数的定义为碰撞点法向碰撞后远离相对速度和碰撞前相对靠近速度.求经过多少次碰撞后杆的最大转角变为原来的$\frac{1}{2}$.



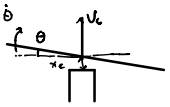


(1) 加速过程

$\ddot{\varphi} = -\frac{mg\frac{l}{2}}{\frac{1}{12}ml^2} = -6\frac{g}{l}$

经过时间t_1后有 $\varphi_0 = \frac{1}{2}(-\ddot{\varphi})t_1^2$

$t_1 = \sqrt{\frac{2\varphi_0 l^2}{6gl}}$;



有角速 $\dot{\theta} = \frac{6g}{l^2}\cdot t_1 = \sqrt{\frac{2\varphi_0 \cdot 6gl}{l^2}}$

$v_c = -\frac{l}{2}\dot{\theta} = -\frac{l}{2}\cdot\sqrt{\frac{2\varphi_0\cdot 6gl}{l^2}}$ 向上为正

此时质心位移 $x=0$ 角度 $\theta=0$

B点碰撞.

$\Delta E=0$: $\frac{1}{2}mv_c^2 + \frac{1}{2}\cdot\frac{1}{12}ml^2\dot{\theta}^2 = \frac{1}{2}mv_c'^2 + \frac{1}{2}\cdot\frac{1}{12}ml^2\dot{\theta}'^2$

B为支. $\Delta L=0$

$mv_c\frac{l}{2} + \frac{1}{12}ml^2\dot{\theta} = mv_c'\frac{l}{2} + \frac{1}{12}ml^2\dot{\theta}'$

改写: $v_c^2 + \frac{l^2}{3l^2}(\dot{\theta}\frac{l}{2})^2 = v_c'^2 + \frac{l^2}{3l^2}(\dot{\theta}'\frac{l}{2})^2$ ①

$v_c + \frac{l^2}{3l^2}(\dot{\theta}\frac{l}{2}) = v_c' + \frac{l^2}{3l^2}(\dot{\theta}'\frac{l}{2})$ ②

令 $x = \theta\frac{l}{2}\frac{L}{gl}$; $\dot{x} = \dot{\theta}\frac{l}{2}\cdot\frac{L}{gl}$ $y = x_c$, $\dot{y} = v_c$

以 x-y 建立坐标.

B点碰撞条件 $x_c = \frac{a}{2}l$;

即 $y = x\cdot\frac{a}{2}$. 令 $\tan\alpha = \frac{a}{2}$; $y = \tan\alpha\cdot x$.

$\Delta E=0$ $\Rightarrow$ 代表点速度大小不变. $\dot{x}^2+\dot{y}^2 = \dot{x}'^2+\dot{y}'^2$

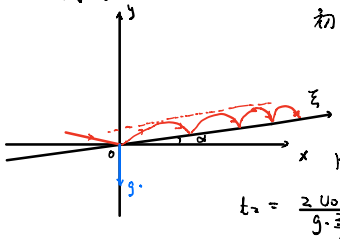
$\Delta L=0$ $\Rightarrow$ $\dot{y} + \frac{l}{gl}\dot{x} = Const$

$\sin\alpha\,\dot{y} + \cos\alpha\,\dot{x} = Const$. $\Rightarrow$ 代表点速度α方向不变

$\Rightarrow$ 代表点在 $y=\tan\alpha\,x$ 线上反射.

非碰撞时 $\ddot{x}=0$. $\ddot{y} = -g$ $\Rightarrow$ 均匀加速.

$\Rightarrow$ 代表点在无滑斜坡上抛体+碰撞.



初态 $\begin{cases}\dot{x} = \cot\alpha\, v_0 \\ \dot{y} = -v_0\end{cases}$

在x方向经过 $t_2 = \frac{2\cdot\frac{v_0}{\sin\alpha}\cdot\cos 2\alpha}{g\cdot\sin\alpha}$

后.还原

$t_2 = \frac{2v_0}{g\cdot\frac{2l}{gl}} = t_1$

$\Rightarrow$ 在 $T = 2(t_1+t_2)$ 后 几乎还原.

(2) $e = 1-\delta$. $\rightarrow$ 相对速度 $\rightarrow (1-\delta)$

能量 $\rightarrow (1-2\delta)$. 最大偏角 $\sim E$

$E \rightarrow \frac{1}{2}E$. $(1-2\delta)^N \simeq \frac{1}{2}$

$N \simeq \frac{\ln 1/2}{\ln(1-2\delta)} \simeq \frac{\ln 2}{2\delta}$.